

Correct answer: $0.75 = \frac{75}{100} = \frac{3}{4}$. So 1.75 metres written as a fraction is $1\frac{3}{4}$ metres.

Standard form

Standard Form (or Standard Index Form) is a way of writing very big or very small numbers in a nice short way by using powers of 10. This is because when you multiply by 10, 100 (10^2) or 1000 (10^3) and so on the effect is that the decimal point moves.

To be technical about it, every number can be written as **$a \times 10^n$** where **a** is a number between 1 and 10 and **n** is an integer (whole number including negative whole numbers)!

For example:

6 000 000 000 can be written as **6×10^9**

0.32 can be written as **3.2×10^{-1}** (Note the **negative** power of 10 because you want the point to move to the **left** and not to the right).

What you need to do:

1. Look at your number written in full and put the decimal point straight after the first non-zero digit.
2. Count how many times it needs to shift to get back to where it was. This gives you your power of 10.

Take this example...

To write 5480000 in standard form, shift the decimal point until it is after the 5, counting how numbers it hopped over on the way. That number is then the number you have to raise 10 to to give you the standard form. **Click 'Play' below to see this in action.**

Now you can put numbers into your calculator that wouldn't even fit on the screen before!

Calculating with standard form

As numbers written in Standard Form are still numbers you can multiply them, divide them, etc. The rules of indices can provide some shortcuts!

For example:

$(2 \times 10^7) \times (4 \times 10^9)$ can be rearranged to $(2 \times 4) \times (10^7 \times 10^9)$ giving 8×10^{16}

Important: The number at the front must be between 1 and 10 so be careful!

You may need to adjust it a bit by moving the point and changing the power of 10 to make up for your change.

For example:

$(5 \times 10^6) \times (7 \times 10^{12}) = (5 \times 7) \times (10^6 \times 10^{12}) = 35 \times 10^{18}$

But the answer is not in Standard Form and the question may ask you to give your answer in Standard Form

Shift the point over the 5 to get 3.5 then increase the power of 10 by **one** to get it back where it was before you moved it. This gives the answer **3.5×10^{19}** .

Exercise 13

In pairs, convert these numbers into standard form.

- | | |
|-------------------|-------------------------|
| 1. 0.000000000467 | – 4.67×10^{-5} |
| 2. 0.0000000316 | - 3.16×10^{-8} |
| 3. 547 | - 5.47×10^2 |
| 4. 0.00000000509 | - 5.09×10^{-9} |
| 5. 4660000000 | - 4.66×10^9 |
| 6. 11000000 | - 1.1×10^7 |
| 7. 0.000703 | - 7.03×10^{-4} |
| 8. 2490000 | - 2.49×10^6 |
| 9. 0.0768 | - 7.68×10^{-2} |
| 10. 54400 | - 5.44×10^4 |